

Sprint 1 Design Document

September 8th - 21st, 2026

Team 4:

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Objective

The primary objective of Sprint 1 is to establish the basic structure and user experience of the EMR (Electronic Medical Records) system. We are implementing the login and user account functionality, instructor dashboard, global search, application navigation, and patient information display. These features establish the primary user flow through the application, from accessing the system and navigating the instructor interface to searching for patients and viewing their medical information.

The sprint will prioritize the primary user views and interactions needed to demonstrate how the overall EMR system will be organized. The instructor dashboard will provide a central interface for viewing important patient and practice information, while global search and the patient information display will establish how users locate and access individual patient records. Representative test patient data will be used to demonstrate patient demographics, medications, lab and test results, and previous visits.

Establishing these core features first will provide a functional foundation for future development while allowing the team to confirm the client's vision for the user interface, information organization, and overall application workflow before expanding into additional EMR functionality.

Features

Feature	Description
Login Page	Establish the application's login process and support instructor, staff, and student user accounts
Instructor Dashboard	Develop the primary instructor interface containing summary information, appointments, activity information, and navigation to available EMR functionality.
Global Search	Provide a global search interface for locating patients and accessing their corresponding patient information

Application Navigation	Establish application routing between the login page, instructor dashboard, search functionality, and patient information pages while restricting protected pages to authenticated users.
Patient Information Page	Develop a centralized patient view containing demographic and medical information, medications, lab and test results, and previous visits. Representative test patient data will be used to develop and demonstrate the interface during Sprint 1.

Technology Stack

The front end will be developed using React.js with TypeScript, with Node.js used for the backend and PostgreSQL used as the relational database. React allows the interface to be divided into reusable components and allows displayed content to be changed based on the user's current application state. JavaScript will be used for most React pages and components, however some supporting files may use TypeScript where appropriate. Node.js with Express.js will be used. Node.js will provide the server-side JavaScript environment while Express.js will provide the framework used to create API endpoints and process requests from the React front end. PostgreSQL will be the relational database as we will need to store strictly structured information like users, patients, appointments, medications, and other EMR information.

This technology stack was selected to support a structured, scalable application while maintaining clear separation between the user interface, backend logic, and data storage.

Functional Requirements

1. Login Page
 - 1.1. The system will provide a login page when a user first opens the application, and it will contain an email address field, password field, and finally a login button.
 - 1.2. The user will be able to enter an email address and password. The system will then validate the submitted information as credentials.
 - 1.3. If the credentials are valid, the system will authenticate the user and display the appropriate dashboard. If the credentials are invalid, then the system will not continue onto the dashboard and display an error message.
 - 1.4. The password field shall hide the entered password from normal view.

- 1.5. After a successful login by the user, the system will determine the authenticated state for the current browser session of the user.
 - 1.6. Refreshing the application during the same browser session should not immediately require the user to login again.
 - 1.7. The system shall provide the user with the option to log out. Then after logging out, the system will remove the active user session and return the user to the login page.
2. Instructor Dashboard
 - 2.1. When an instructor successfully logs in, the system will display the instructor dashboard and identify the currently authenticated instructor.
 - 2.2. The dashboard will display summary information representing important activity within the EMR system that either the patient or user has interacted with. The Sprint 1 dashboard may use demonstration data because complete backend functionality is outside the scope for this sprint.
 - 2.3. The dashboard will provide summarization cards for total patients, Today's appointments, Pending tasks, and Patient or practice activity
 - 2.4. The dashboard will display a section containing scheduled appointments, visual representation of weekly system activity, and the interface shall provide navigation controls that establish additional EMR functions.
3. Global Search
 - 3.1. The system shall provide a global search function accessible from the primary application interface.
 - 3.2. The global search shall allow an authenticated user to search for patients using available identifying information.
 - 3.3. The system shall display patients matching the provided search criteria.
 - 3.4. Search results shall provide sufficient identifying information for the user to distinguish between matching patients.
 - 3.5. Selecting a patient from the search results shall provide access to the corresponding patient information page.
 4. Application Navigation
 - 4.1. Unauthenticated users will be shown the login page. The application will then determine whether a user is authenticated before displaying the dashboard.
 - 4.2. Authenticated administrators will be shown the administrator interface.
 - 4.3. Additional page routing will be expanded in later sprints.

5. Patient Information Display

- 5.1. When an administrator selects a patient, the system shall display a patient information page associated with the selected patient.
- 5.2. The patient information page shall display the patient's identifying and demographic information, including information such as the patient's name, date of birth, contact information, and other relevant patient details.
- 5.3. The page shall provide access to the patient's medical information, including current and past medications, medical history, and other relevant medical records.
- 5.4. The page shall display references to laboratory and test results associated with the patient. These records shall identify the type of test or laboratory work, the date performed, and the corresponding result or status when available.
- 5.5. The page shall provide a record of the patient's previous visits. Past visit information shall include relevant details such as the visit date, reason for the visit, and associated visit summary.
- 5.6. Medical information, laboratory and test results, medications, and visit records displayed on the page shall correspond to the currently selected patient.

Implementation

1. Login Page

The login page will be implemented as a React component containing controlled input fields for the user's email address and password. User accounts will include a role identifying the user as an instructor, staff member, or student. Authentication state will be maintained for the browser session and used to control access to protected application pages. Logging out will clear the active session and return the user to the login page.

2. Instructor Dashboard

The instructor dashboard will be implemented using reusable React components for summary cards, appointments, system activity, and navigation controls. Dashboard components will be structured so that demonstration data used during Sprint 1 can later be replaced with information retrieved from backend API endpoints.

3. Global Search

Global search will be implemented as a reusable React component that

submits patient search criteria to the Node.js/Express backend. The backend will query available patient records and return matching results to the front end. Each result will reference a unique patient identifier that can be used to access the corresponding patient information page.

4. Application Navigation

Application navigation will be implemented using React Router to define routes between the login page, instructor dashboard, and patient information display.

5. Patient Information Display

- 5.1. The patient information page will be implemented as a React view composed of sections for patient demographics, medical information, medications, laboratory and test results, and past visits. Each patient will be identified by a unique patient ID, which will be used by the backend to retrieve records associated with that patient.